

Tendons, Ligaments and Injuries

COL5A1, COL1A1, COL3A1, GDF5, muscle disease/ligation/fascia scores, shoulder injuries, Dupuytren and arthrosis, and tendon loci load and lesions

Sérgio F · 03-10-2026 · local time

Informative and educational genetic report, based only on Sérgio F genes (genotyping chip + local imputation). It is neither a diagnosis nor a prescription. All actions are marked as suggestion (not applied).

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1. Summary

Favourable. The genomic scores of soft tissues are typical to low: muscle diseases, ligament and fascia in percentile 34.1, other soft tissue diseases in the 24.6 and Dupuytren in the 25.1. It does not have the low expression allele of **GDF5** (G/G; consistent with the favourable GDF5 of §27) or the COL3A1 Ala698Thr. The load of tendon loci, ligaments and shoulder is in percentile 57.1, the carpal tunnel in the 6.3 and Dupuytren in the 16.3.

Unfavourable. COL5A1 rs12722 is T/T (2 copies of the allele associated with Achilles tendinopathy and anterior cruciate ligament rupture, already mentioned in §27). COL1A1 does not have the protective Sp1 allele. Shoulder injuries (rotator cuff, capsulitis) are in percentile 56.9 and knee arthrosis in the 56.1: typical, slightly above the median.

What does it mean? The genetics of tendons and ligaments is globally typical. There is only one unfavourable classical sign (COL5A1 T/T, of weak evidence), without confirmation in the genomic scores. The real risk comes mainly from the progression of the load: the tendon adapts slower than the muscle, and this weighs more with strength training as a priority (Reports Sports Profile and §27). The spine and disc are in §50; the bone and the general arthrosis in §27.

2. How to read this report

- **Data:** Sérgio F chip genotypes (GRCh38, strand +) and, when SNP is not in the chip, locally imputed genotypes (Beagle 5.5, public panel of genomes complete to 30x) with quality $DR2 \geq 0.8$. Under this threshold SNP is treated as SNP not found in the data.
- **Keyvariants** 4 Selected SNPs from the literature (GWAS Catalog, ClinVar, dbSNP and reference studies): 4 of the chip, 0 imputed and 0 not found.
- **Strand:** All genotypes are in the GRCh38 strand +. The published effect allele is compared with the observed alleles; 0 SNPs needed conversion to the complementary strand and 0 are palindromic (A/T or C/G), solved by the reference allele and the position. When the literature uses the strand of the gene, the letters may differ from those of this report.
- **Polygenic scores (PGS):** 6 PGS Catalog scores calculated locally with 5.556.727 variants in total (chip + imputed). The percentile compares Sérgio F with the reference population ($n = 633$) calculated with exactly the same variants. Percentage ≥ 80 : MAIS PROVABLE/ALTO; ≤ 20 : MENOS PROVABLE/BAIXO; between 20 and 80: TIPIC.
- **Loci load GWAS:** For each trait, all SNPs with $p \leq 5 \times 10^{-8}$ in GWAS Catalog (1 line for SNP, the most significant) are added as simple count of effect alleles (each allele valley 1). The sum of Sérgio F is compared with that of the reference population ($n = 633$) with exactly the same variants, which gives an empirical percentile. Many of these SNPs are in linkage disequilibrium, so the load is a complement, not a substitute, of the polygenic scores. 733 loci, 659 with genotype.
- **Judgment:** Each result is divided into Favourable or Unfavourable (or typical/without judgment). A genotype is a probability, not a destination; almost all effects are small and add to the lifestyle.
- **Evidence:** Strong : replicated and with clear clinical effect; Moderate = GWAS/consistent meta-analysis, modest effect; Weakness : small or mechanistic studies.

3. Polygenic scores

High percentile : higher risk. The scores of soft tissue diseases group clinical diagnoses (tendinitis, enthesopathies, fasciitis, shoulder injuries).

Soft tissue

Trait	Score (PGS Catalog)	Variants used	Percentile	Result	Judgement
Muscle, ligament and fascia disorders	PGS019710 (Chen 2025, PRS-CS, 984 thousand variants)	950.693 983.801 (weighted coverage 94%)	34,1	TIPIC	typical
Shoulder injuries (rotator cuff, capsulitis, tendinitis)	PGS004488 (Jung 2024, LDpred2, 1.06 million variants)	1.036.079 1.059.939 (weighted coverage 97%)	56,9	TIPIC	typical
Other diseases of soft tissues	PGS004489 (Jung 2024, LDpred2, 1.06 million variants)	1.036.079 1.059.939 (weighted coverage 97%)	24,6	TIPIC	typical
Dupuytren's disease	PGS002092 (Privé 2022, LDpred2, 484 thousand variants)	475.662 484.484 (weighted coverage 97%)	25,1	TIPIC	typical

Articulations

Trait	Score (PGS Catalog)	Variants used	Percentile	Result	Judgement
Knee arthrosis	PGS002767 (Mars 2022, PRS-CS, 1.05 million variants)	1.029.107 1.052.275 (weighted coverage 97%)	56,1	TIPIC	typical
Hip arthrosis	PGS002763 (Mars 2022, PRS-CS, 1.05 million variants)	1.029.107 1.052.273 (weighted coverage 97%)	38,9	TIPIC	typical

4. Key variants by topic

Collagen and GDF5 variants studied in sports injuries. The evidence is mainly small case-control studies; the effect is small.

Structural collagens

Tendons and ligaments are mainly made of collagen I, with collagen III and V regulating the diameter of fibrils. Variants in these genes are associated, in case-control studies, with tendinopathy and ruptures.

Gene	SNP	Variant	Genotype	Source	Effect allele (copies)	Judgement	Allele effect	Evidence
COL5A1	rs12722		T/T	chip	T (2 copy/s)	unfavourable	BstUI (3'UTR): T allele is associated with Achilles tendinopathy, anterior cruciate ligament rupture and muscle cramps (case-control studies; §27 detail)	Weak
COL1A1	rs1800012	Sp1	C/C	chip	A (0 copy/s)	neutral	Sp1 (G>T in the gene; A in strand +): allele A (T) is associated with less ruptures of the anterior cruciate ligament and shoulder dislocation	Weak
COL3A1	rs1800255	A698T	G/G	chip	A (0 copy/s)	favourable	Ala698Thr: Allele A is associated with more ruptures of the anterior cruciate ligament in some studies (detail in §28)	Weak

Cartilage and signaling

GDF5 is a growth factor of cartilage and tendon; the low expression allele is associated with arthrosis and tendinopathy.

Gene	SNP	Variant	Genotype	Source	Effect allele (copies)	Judgement	Allele effect	Evidence
GDF5	rs143383		G/G	chip	A (0 copy/s)	favourable	+104T/C (5'UTR): A allele (T in the gene) lowers GDF5 expression (cartilage and tendon growth factor); associated with Achilles arthrosis and tendinopathy	Moderate

5. GWAS loci burden

Sum of all loci of GWAS Catalog ($p \leq 5 \times 10^{-8}$) of tendon and ligament lesions (Aquiles, rotator cuff, frozen shoulder, sprains), carpal tunnel, Dupuytren and arthrosis, with alleles oriented towards more risk, compared to the reference population.

Trait (GWAS Catalog)	Loci (with genotype)	Sérgio F effect alleles	Calibrated variants	Percentile	Judgement
Tendons, ligaments and shoulder	38 (17)	14 de 34	17	57,1	typical
Carpal tunnel syndrome	68 (63)	56 de 126	63	6,3	favourable
Dupuytren's disease	126 (116)	83 de 232	116	16,3	favourable
Knee and hip arthrosis	501 (463)	444 de 926	462	22,1	typical

6. Favourable

- **Tissues typical of low:** muscle diseases, ligament and fascia in percentile 34.1; other soft tissue diseases in the 24.6. **Moderate.**
- **Dupuytren** in percentile 25.1 (loci load in 16.3). **Moderate.**
- **GDF5 rs143383 G/G:** without the low expression allele (arthritis, tendinopathy). **Moderate.**
- **COL3A1 Ala698Thr absent** (G/G). **Weak.**
- **Hip arthrosis** in percentile 38.9; knee and hip loci load in the 22.1. **Moderate.**
- **Tendons, ligaments and shoulder:** loci load in 57.1; carpal tunnel in 6.3. **Weak.**

7. Unfavourable

- **COL5A1 rs12722 T/T**: 2 copies of the allele associated with Achilles tendinopathy, rupture of the anterior cruciate ligament and cramps associated with exercise (OR ~1.5–2 in small studies; not confirmed in GWAS). **Weak.**
- **COL1A1 Sp1 C/C**: without the protective allele (TT in the gene) of ligament ruptures; is the most common genotype. **Weak.**
- **Shoulder injuries** in percentile 56.9 and knee arthrosis in the 56.1: typical, slightly above the median. **Weak.**

8. Mechanisms of ageing

Mapping in 84 mechanisms of aging of Sérgio F immortality objectives (only the mechanisms affected by this report).

Mechanism	Genetic finding	Meaning
Remodelling of the extracellular matrix	COL5A1 T/T (diameter of collagen fibrils) against GDF5 and COL3A1 favourable	Mixed
Stiffness of the Extracellular Matrix	Tendency collagen with age and glucose (see §28); typical genetics	Neutral
Chronic Inflammation	Tendinopathy and enthesopathy: soft tissue scores typical to low	Neutral to favourable
Exhaustion of Staminary Cells	The tendon progenitors decrease with age, so the adaptation is slower (without specific genetic marker)	Neutral
Fragility	Arthritis of the typical knee and hip to low; Dupuytren	Neutral to favourable

9. Practical suggestions

All suggestions are suggestion (not applied), consistent with the current genetic plan, and should be discussed with the doctor.

1. **Gradual load progression:** increase the volume or intensity ~10%/week, and not both at the same time. Tendons need weeks to months to follow the muscle. *Suggestion (not applied).*
2. **Tendency training:** slow and heavy exercises (3 s to go up, 3 s to go down) and eccentric exercises for Achilles, rotulian and rotator coiffus, 2×/week, as prevention. *Suggestion (not applied).*
3. **Hydrolyzed collagen or gelatin 10-15 g + vitamin C ~50 mg, 30-60 min before tendon training** (as in §27). *Suggestion (not applied).*
4. **Warming and mobility of the shoulder** before the training of upper limbs (shoulder injuries in the typical-high percentile); avoid suddenly rising the load above the head. *Suggestion (not applied).*
5. **Drugs with tendon risk:** fluoroquinolones (cyprofloxacin, levofloxacin) increase the risk of tendon rupture, even more with corticosteroids. To ask for an alternative to the doctor whenever possible; if necessary, reduce the load during treatment and in the following weeks. Avoid repeated injections of corticosteroids in the tendons. *Suggestion (not applied).*
6. **Isotretinoin:** can cause muscle and joint pain and rise to CK. During the treatment, adjust the load and not interpret the high CK without this context. *Suggestion (not applied).*
7. **Alert signs:** pain in the tendon that worsens during the session or the following morning means reducing the 1-2 load weeks and not stopping completely (the controlled load is treatment). *Suggestion (not applied).*
8. **Low glucose and glycation** (Glycosis report; §28: glycation makes the collagen of the tendon more rigid. *Suggestion (not applied).*

10. Limitations

- The genetics of tendon and ligament lesions is weak: candidate SNPs (COL5A1, COL1A1, COL3A1) come from small case-control studies and were not confirmed in large GWAS.
- The soft tissue scores cluster different clinical diagnoses; there are no specific scores of anterior cruciate ligament rupture or Achilles.
- The Achilles GWAS, rotator cuff and frozen shoulder have few loci; the loci load is descriptive.
- Conclusions only from genes; it is not a diagnosis.

11. Methods and sources

- **Genotypes:** Sérgio F chip (GRCh38, strand +). Absent SNPs come from local imputation (Beagle 5.5, public panel of complete 30x genomes), used only with $DR2 \geq 0.8$. No genetic data came out of the analysis computer.
- **Alleles and strand:** The published allele of effect is compared with the alleles of the Imposed VCF; in the palindromics the strand + (standard of the GWAS Catalog and the PGS Catalog) is assumed.
- **Polygenic scores:** GRCh38 files from PGS Catalog (6 scores: muscle, ligament and fascia diseases, shoulder injuries, other soft tissue diseases, Dupuytren and knee and hip arthrosis). Score = Δ coefficient $\times$ dosage. The percentile is empirical in the reference population ($n = 633$), with the same variants.
- **Loci load:** GWAS Catalog ($p \leq 5 \times 10^{-8}$), alleles guided by the published effect signal, and empirical calibration in the same reference population.
- **Interpretation:** reference literature (September 2009 and Posthumus 2009 for COL5A1; Posthumus 2009 for COL1A1; St?? Posthumus 2010 for GDF5; Kim 2021 for the GWAS of lesions Bisset 2015 for drugs and tendons, GWAS Catalog and ClinVar. The complete references are available on request.

12. Annex: all variants analysed

Topic	Gene	SNP	Genotype	Source	Effect allele	Copies	Chain	Interpretation
Structural collagens	COL5A1	rs12722	T/T	chip	T	2	strand +	2 risk allele copies
Structural collagens	COL1A1	rs1800012	C/C	chip	A	0	strand +	without the protective allele (common genotype)
Structural collagens	COL3A1	rs1800255	G/G	chip	A	0	strand +	without the risk allele
Cartilage and signaling	GDF5	rs143383	G/G	chip	A	0	strand +	without the risk allele